

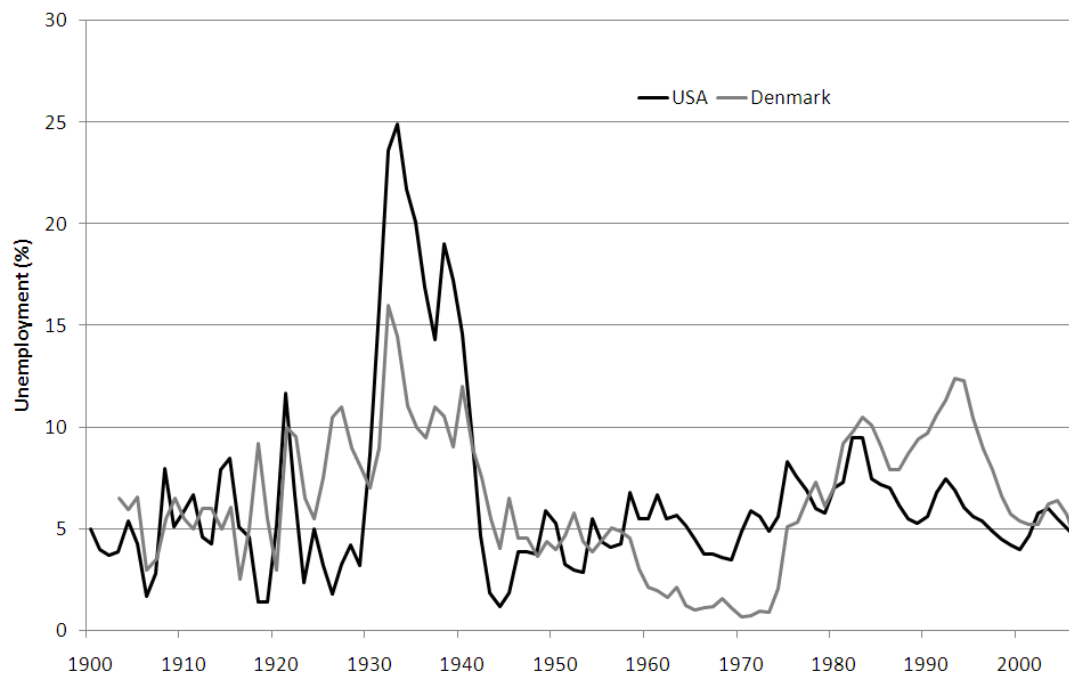
Chapter 10

Macroeconomics III
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SOME FACTS AND
INTRODUCTORY
THEORY ABOUT
UNEMPLOYMENT

STRUCTURAL UNEMPLOYMENT

- In our growth models, L_t = supply of labour = demand of labour = employment. No unemployment.
- Rate of unemployment, annual data, USA and Denmark:



- Long run: no trend, "natural" average level around 5-7%
- Short run: fluctuations around natural rate, sometimes large

- Growth theory is about the long run: we can interpret the L_t of our models as, e.g., 95% of the labour force and let the models be otherwise unchanged.
- The natural level of unemployment is of great interest for society (e.g., is it 4% or 8%?) among other reasons because:
- **Involuntary unemployment implies a social waste:**
 - Involuntary unemployment $\Rightarrow$ there are unemployed workers for whom:

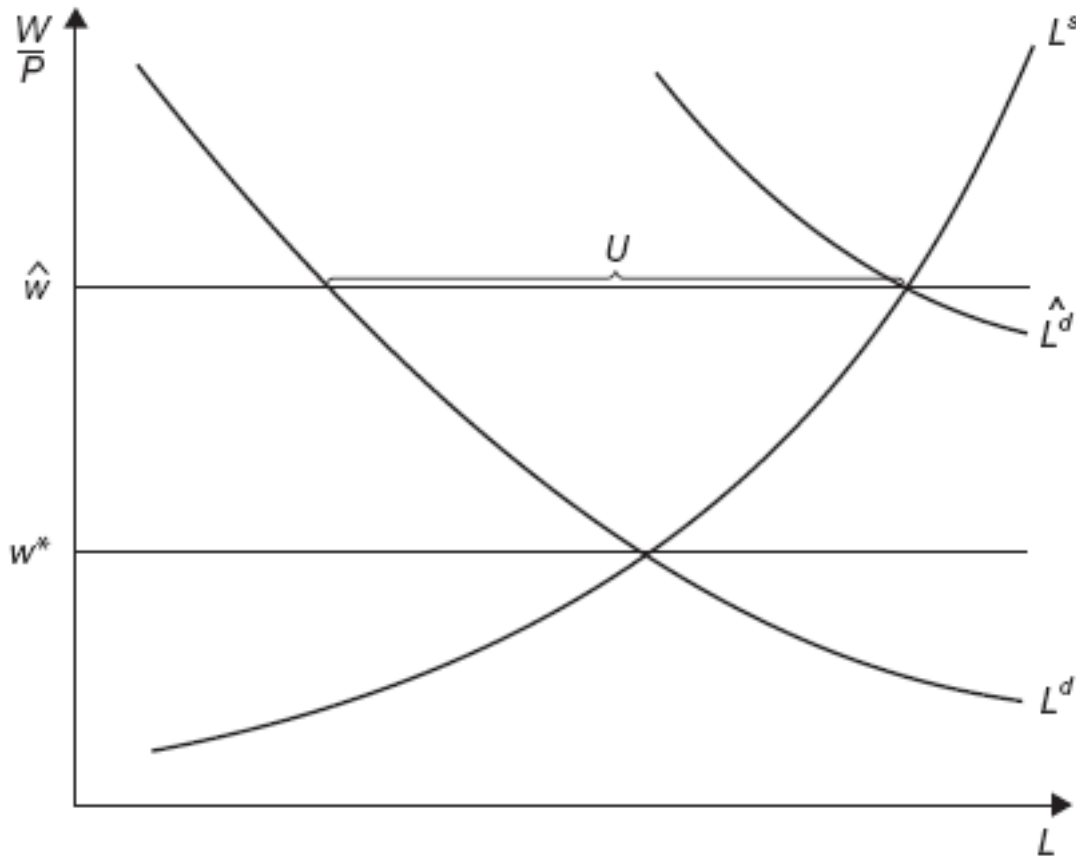
$$MRS < W / P.$$
 - Profit maximization $\Rightarrow$ for the marginally employed it must hold that

$$W / P \leq MP.$$
 - That is, involuntary unemployment + profit maximization $\Rightarrow$ there are unemployed for whom:

$$MRS < MP.$$
 - But then there is a social waste! Putting the marginally unemployed to work produces more additional output than needed to compensate.
- There are different theories for explaining long-run and short-run unemployment.
- Now: structural, long-run, natural unemployment

CYCLICAL AND STRUCTURAL UNEMPLOYMENT

- Involuntary unemployment is by definition **almost** the same as excess supply of labour. Consider a labour market:



- Unemployment: the wage is stuck at a "too high" level. Shouldn't this imply a downward pressure on wages which would in itself eliminate unemployment? This is the key question in all theory of unemployment.

- Under perfectly competitive wage flexibility (by definition meaning that the nominal wage rate always adjusts to create a real wage rate that equates supply and demand of labour), there can be no unemployment.
- In other words:

For unemployment to exist, some factor must have prevented the real wage from adjusting to a level that would equate (wage taking) supply and demand of labour.
- Such a factor is called a "wage rigidity".
- **For unemployment to prevail, there must be wage rigidity.**
- Why and how does wage rigidity arise?

TYPES OF WAGE RIGIDITY

Short-run nominal wage rigidity and cyclical unemployment

- The unemployment does in fact create a downward pressure on the nominal wage rate, W , (and hence on the real wage rate, W / P), but the adjustment takes time, e.g., because nominal wages are negotiated one period ahead. There can then be a period where the nominal wage rate does not have time to adjust and this can create unemployment: **short-run or cyclical unemployment.**
- As time goes by, cyclical unemployment should disappear by market forces, but **this could take a long time** and **new shocks are occurring all the time.**
- The real wage rate, W / P , can be flexible under short-run nominal wage rigidity because of adjustments in P , but not in a way that would eliminate unemployment because P is not determined in the labour market.

LONG-RUN REAL WAGE RIGIDITY AND STRUCTURAL UNEMPLOYMENT

- Assume that there were no new shocks and that the time horizon is long enough to make nominal wage rigidities irrelevant. How can there be unemployment?
- Only if unemployment for some reason does **not** imply a downward pressure on the real wage rate. Such a reason is called a long-run, real wage rigidity.
- Long-run, real wage rigidity implies that the economy's general equilibrium system establishing the relative/real prices cannot be characterized by perfect competition.
- Unemployment arising from long-run, real wage rigidity is called **long-run, structural or natural unemployment**.
- Long-run, real wage rigidity is a factor that can explain how unemployment and fully adjusted wages and prices can coexist.

CAUSES OF STRUCTURAL UNEMPLOYMENT

$\cong$ causes of long-run wage rigidity $\cong$ explanations of the coexistence of unemployment and adjusted wages:

- How can we have an unemployment rate of 5-7% and no downward pressure on wages?
- **Frictional unemployment:** it takes time to move from one job to another.
- **Search unemployment:** there is uncertainty about job possibilities, and therefore the unemployed choose to remain unemployed for some time while searching for jobs.
- **Efficiency wages:** the labour market is basically competitive, but labour's productivity depends on wage.
- **Trade unions:** the labour market is not competitive.

ALL UNEMPLOYMENT

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graph TD; A[ALL UNEMPLOYMENT] --> B[Short-run or cyclical unemployment]; A --> C[Long-run or structural or natural unemployment];
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Short-run or cyclical unemployment

REASON:

Short-run nominal wage rigidity

POLICY:

STABILIZATION POLICIES,
e.g. fiscal and monetary policy

Long-run or structural or natural unemployment

REASON:

Long-run real wage rigidity

SUBDIVISION by cause of rigidity:

- **FRICTIONAL** unemployment (involuntary)
- **SEARCH** unemployment (voluntary)
- Unemployment due to **EFFICIENCY WAGES** (involuntary)
- Unemployment due to **TRADE UNIONS** (involuntary/voluntary)

POLICY:

STRUCTURAL (LABOUR MARKET) POLICIES,
e.g. reforms of the unemployment benefit system, the educational system, or the tax system.